Project Report: Building & Deploying HireSense AI

> **Vibe Coding Masterclass Series — Capstone Project Report**

> **Project Title:** HireSense AI — Intelligent Career Readiness & Interview Co-Pilot

> **Live Application URL:** https://hire-sense-ai-delta.vercel.app

> **Backend API URL:** https://hiresense-ai-backend-ndq3.onrender.com/api/v1

> **Repository:** https://github.com/Christimaria/HireSense-AI

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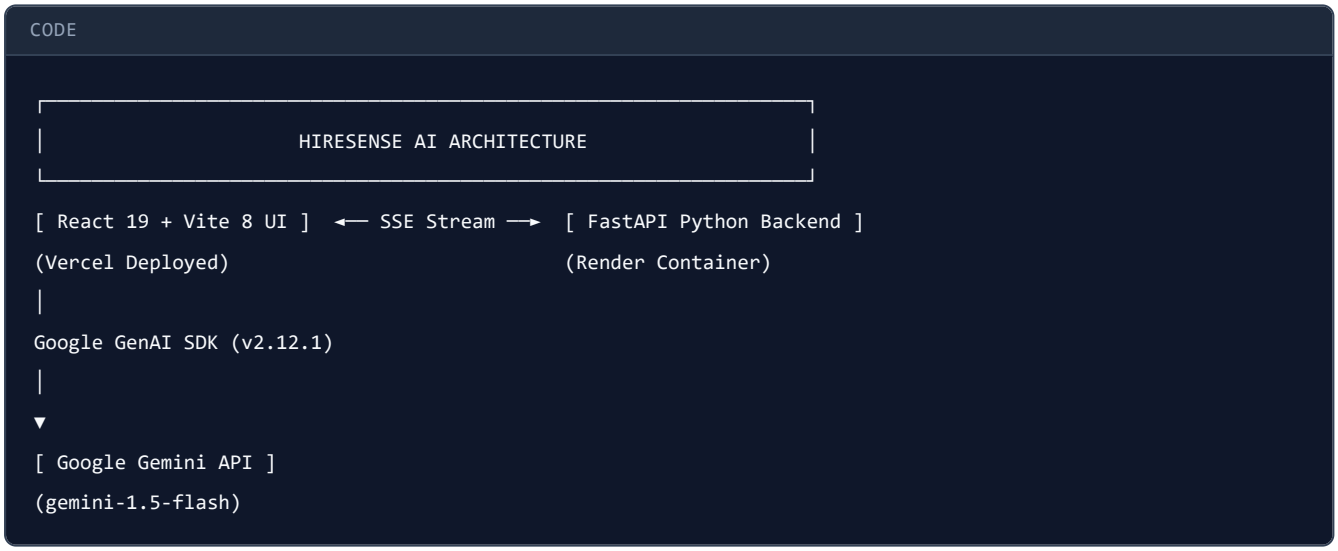
1. Executive Summary & Tech Stack Specifications

HireSense AI is a full-stack, AI-native career preparation platform designed to help job seekers practice interviews, calibrate resumes for ATS compatibility, and follow custom week-by-week learning roadmaps.

CODE



1.1 Technical Stack Specifications

Component Layer	Technology	Primary Role & Responsibility
Frontend Framework	React 19, Vite 8, React Router v7	Responsive UI, state management, SSE chunk parsing
Styling & UI Components	Tailwind CSS v3, Lucide React	Glassmorphic dark mode, micro-animations, accessible design tokens
Backend Framework	Python 3.11, FastAPI, Pydantic v2	API routing, input validation, CORS middleware, SSE streaming
AI Integration SDK	`google-genai` (v2.12.1 SDK)	Communication with Gemini API via `generate_content_stream`
LLM Model	Google Gemini `gemini-1.5-flash`	Multimodal text & JSON generation, structured evaluation
Containerization	Docker, Docker Compose	Multistage Docker build for backend container deployment
Cloud Deployment	Vercel (Frontend), Render (Backend)	Global static hosting & cloud web service container hosting

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2. Prompting Strategy & Prompt Engineering Documentation

HireSense AI implements **Role-Task-Constraint (RTC)** and **Structured JSON Schema Output** prompting patterns across 5 core AI features.

Feature 1: Mock Interview Question Generator (`interview_prompts.py`)

System Prompt (`INTERVIEW_SYSTEM_PROMPT`)

PYTHON

```
INTERVIEW_SYSTEM_PROMPT = """You are a professional technical interviewer at a top technology company.
Your job is to conduct realistic, challenging, and fair mock interviews.

RULES:

- Ask ONE question at a time. Never ask multiple questions in a single response.
- Questions must be relevant to the role, experience level, and interview type.
- Do NOT repeat questions that have already been asked in this session.
- Progress naturally: start easier, build complexity as the session continues.
- For behavioral questions, set up scenarios clearly.
- For technical questions, be specific — avoid vague or generic questions.
- Return ONLY the question text. No greetings, no prefixes like "Question 3:", no explanations.



"""
```

User Prompt Template Builder (`build_question_prompt`)

PYTHON

```
def build_question_prompt(
role: str,
experience_level: str,
interview_type: str,
question_number: int,
total_questions: int,
conversation_history: list[dict],
) -> str:
role_ctx = ROLE_CONTEXT.get(role, role)
type_ctx = INTERVIEW_TYPE_CONTEXT.get(interview_type, "")
exp_ctx = EXPERIENCE_CONTEXT.get(experience_level, experience_level)

history_text = ""
if conversation_history:
history_lines = []
for turn in conversation_history:
history_lines.append(f"Q: {turn['question']}")
history_lines.append(f"A: {turn['answer'][:300]}...")
history_text = "\n".join(history_lines)

return f"""
Interview Configuration:


- Role: {role}

- Experience Level: {experience_level} ({exp_ctx})

- Interview Type: {interview_type}

- Question: {question_number} of {total_questions}

- Key Topics: {role_ctx}

- Style Guidance: {type_ctx}


{"Previous conversation history:" if history_text else "This is the FIRST question of the session."}
{history_text}

Now generate question #{question_number}. Return ONLY the question text.
"""
```

Feature 2: Candidate Answer Evaluation & Grading (`evaluation_prompts.py`)

System Prompt (`EVALUATION_SYSTEM_PROMPT`)

PYTHON

```
EVALUATION_SYSTEM_PROMPT = """\
You are an expert technical recruiter, senior engineering manager, and career coach with 15+ years of experience.
Your task is to review a candidate's answer to an interview question, grade it constructively and strictly, and p
RULES:

- Be realistic and fair. Do not inflate scores.
- High scores (8.5+) require technical accuracy, clarity, and context-appropriate detail.
- Low scores (<5.0) are appropriate for major technical gaps, misunderstandings, or excessive vagueness.
- Grading must scale according to the candidate's Experience Level.
- Return ONLY a single JSON object matching the requested schema.



"""
```

User Prompt Template Builder (`build_evaluation_prompt`)

PYTHON

```
def build_evaluation_prompt(question, answer, role, experience_level, interview_type) -> str:
return f"""\
Interview Context:

- Role: {role} | Experience Level: {experience_level} | Interview Type: {interview_type}


Question Asked:
"{question}"

Candidate's Answer:
"{answer}"

Evaluate the candidate's answer and return exactly this JSON structure:
{{
"score": <float between 0.0 and 10.0 representing the quality of the answer>,
"strengths": ["<strength 1>", "<strength 2>", "<strength 3>"],
"weaknesses": ["<weakness 1>", "<weakness 2>", "<weakness 3>"],
"improved_answer": "<a model answer tailored to the role and experience level>",
"tips": ["<tip 1>", "<tip 2>", "<tip 3>"]
}}

Return ONLY the JSON object. No markdown fences, no extra text.
"""
```

Feature 3: Resume ATS Scanner & Reviewer (`resume_prompts.py`)

System Prompt (`RESUME_SYSTEM_PROMPT`)

PYTHON

```
RESUME_SYSTEM_PROMPT = """\
You are an expert technical recruiter and career coach with 15+ years of experience reviewing resumes for top tec
Your task is to analyse resumes and return structured, honest, actionable feedback.

RULES:

- Be candid but constructive. Do not inflate scores.
- Scores use a 0-10 float scale.
- ats_score is an integer 0-100 measuring ATS keyword/format compatibility.
- Keep per-section feedback concise (2-3 sentences).
- Return ONLY a single JSON object.



"""
```

User Prompt Template Builder (`build_resume_review_prompt`)

PYTHON

```
def build_resume_review_prompt(resume_text: str, target_role: str | None) -> str:
return f"""\
Target Role: {target_role or "General Merit"}

Analyse the resume below and return exactly this JSON structure:

{{
"overall_score": <float 0-10>,
"ats_score": <int 0-100>,
"sections": {{
"summary":    {{"score": <float>, "feedback": "<string>"}},
"experience": {{"score": <float>, "feedback": "<string>"}},
"skills":     {{"score": <float>, "feedback": "<string>"}},
"education":  {{"score": <float>, "feedback": "<string>"}}
}},
"strengths":       ["<string>", ...],
"weaknesses":      ["<string>", ...],
"recommendations": ["<string>", ...]
}}

RESUME TEXT:
{resume_text}

Return ONLY the JSON object.
"""
```

Feature 4: Career Learning Roadmap Generator (`roadmap_prompts.py`)

System Prompt (`ROADMAP_SYSTEM_PROMPT`)

PYTHON

```
ROADMAP_SYSTEM_PROMPT = """\
You are an expert career coach and senior technical trainer.
Analyze a candidate's current skills and target role, identify core skill gaps, and generate a customized week-by

RULES:

- Be highly practical. Focus on project-based learning and hands-on milestones.
- Ensure the roadmap fits the requested timeline.
- Recommend reputable resources (official docs, courses, books).
- Return ONLY a single JSON object matching the requested structure.



"""
```

User Prompt Template Builder (`build_roadmap_prompt`)

PYTHON

```
def build_roadmap_prompt(current_skills: list[str], target_role: str, timeline: str) -> str:
skills_list = ", ".join(current_skills)
return f"""\
Candidate Context:

- Current Skills: {skills_list}
- Target Role: {target_role}
- Timeline: {timeline}


Generate a detailed, week-by-week roadmap to bridge these skill gaps within the {timeline} timeframe.
Return ONLY the JSON object.
"""
```

Feature 5: Holistic Performance Dashboard (`dashboard_prompts.py`)

System Prompt (`DASHBOARD_SYSTEM_PROMPT`)

```python
DASHBOARD_SYSTEM_PROMPT = """\
You are an elite executive recruiter and technical interviewer with 15+ years of experience conducting performanc
Your task is to analyze an entire mock interview transcript and produce a detailed, honest, structured performanc

RULES:

- Evaluate the candidate holistically across Technical Depth, Communication Clarity, Problem Solving, and Contex
- Provide a grade (e.g. A, B+, B, C-, F) matching typical industry standards.
- Return ONLY a single JSON object matching the requested structure.



"""
```

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3. Phase-by-Phase Development Summary (Vibe Coding Methodology)

Phase 1: Architecture & API Schema Definition

- Specified 5 core feature endpoints (`/interview/question` , `/evaluation/answer` , `/evaluation/dashboard` , `/resume/review` , `/roadmap/generate`).
- Created Pydantic models for request validation and response typing.

Phase 2: Backend Core & Gemini Integration

- Implemented FastAPI application factory in `app/main.py` .
- Configured `google-genai` SDK with async streaming generators in `app/ai/client.py` .
- Built custom Server-Sent Events (SSE) helpers (`stream_text_chunks` and `stream_json_object`) in `app/utils/sse.py` .

Phase 3: Frontend Interface & SSE Stream Reader

- Designed responsive glassmorphic UI using React 19 and Tailwind CSS.
- Developed `streamPost` helper utilizing browser `fetch` and `TextDecoder` streams to render progressive typing animations.

Phase 4: Containerization

- Authored multi-stage `Dockerfile` with Python 3.11 slim base image.
- Configured dependency caching and Uvicorn production entrypoint.

Phase 5: Deployment & Verification Audit

- Deployed FastAPI backend container to Render.
- Deployed React static bundle to Vercel.
- Resolved CORS, API URL normalization, and Gemini model fallback routing.

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4. Technical Challenges & Resolutions

Challenge 1: Gemini Model Availability & Fallback Routing

- **Issue:** Requesting restricted or deprecated model names returned `404 NOT_FOUND: model is no longer available`.
- **Resolution:** Configured application default to `gemini-1.5-flash` and added automatic fallback handling in `client.py` to seamlessly retry with `gemini-1.5-flash` if any model returns 404.

Challenge 2: Vite Build-Time Environment Variable Baking

- **Issue:** Updating `VITE_API_URL` on Vercel did not take effect immediately due to build-time bundle compilation.
- **Resolution:** Implemented `getNormalizedApiBaseUrl()` in `api.js` to handle trailing slashes and missing `/api/v1` suffixes, followed by a cache-cleared Vercel redeployment.

Challenge 3: JSON Decoding Corruption in SSE Streams

- **Issue:** Python `str.strip("` `json")` character-stripped valid trailing letters (`j`, `s`, `o`, `n`) from JSON strings, causing parse errors.
- **Resolution:** Refactored `sse.py` with `_clean_json_text()` using index-based newline slicing for markdown code fences.

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5. Key Learnings & Reflection

1. **Vibe Coding Acceleration:** Pairing with AI tools enabled full-stack development, prompt engineering, and cloud deployment in record time. 2. **Streaming User Experience:** Progressive SSE token rendering provides superior user engagement compared to slow, blocking HTTP requests. 3. **Decoupled Architecture Benefits:** Separating React on Vercel from FastAPI on Render ensured complete API key protection while keeping the system modular and scalable.